

Financial Market Movement Prediction Using News Sentiment and Market Data

An end-to-end Apple (AAPL) data science project combining market indicators, financial-news NLP, classification models, chronological evaluation, SHAP explainability, and an interactive Streamlit dashboard.

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Asset	Apple Inc. (AAPL)
Prediction target	Whether the next trading day's closing price increases
Evaluation period	7 Aug 2023 to 27 Nov 2024 (332 trading days)
Report date	August 2026

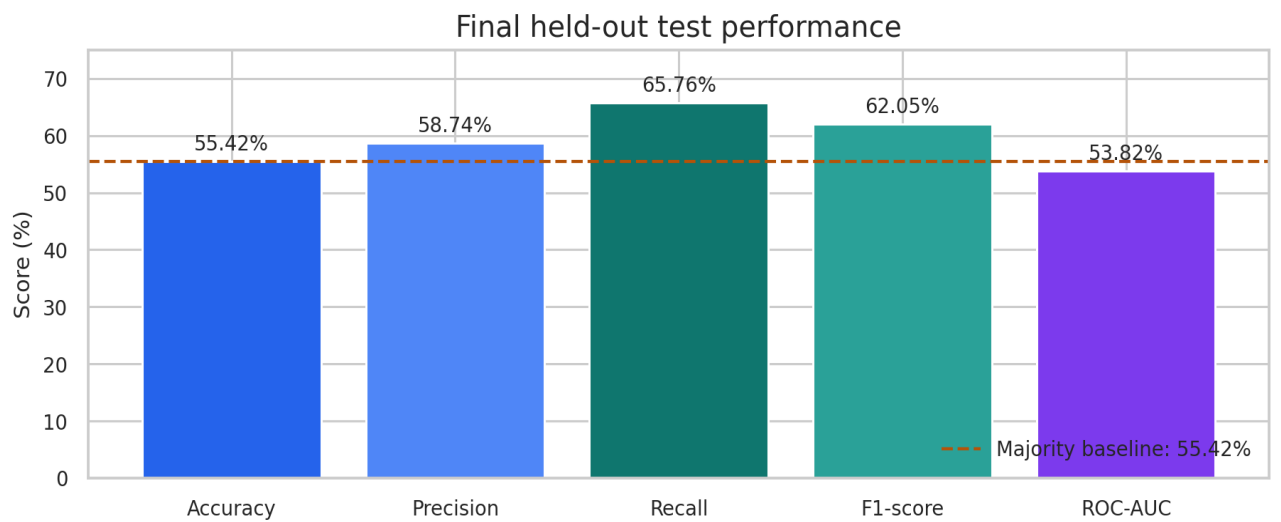
Live application: raginiboya-financial-market-prediction.streamlit.app

Source repository: github.com/raginiboya/financial-market-prediction

Executive Summary

This project built and evaluated a next-trading-day direction classifier for AAPL. It integrated historical price and volume data, technical indicators, a supervised financial-headline sentiment pipeline, daily news aggregates, four candidate classifiers, chronological train-validation-test splitting, and SHAP-based model interpretation.

The locked final model was a market-only logistic regression selected using validation diagnostics. On the untouched test set it achieved 55.42% accuracy and 53.82% ROC-AUC. Accuracy equaled the test majority-class baseline, while the model achieved 65.76% recall and 62.05% F1-score for upward days. These results indicate weak directional signal rather than production-grade predictive power.



Unified observations	2,210 trading days	2016-02-19 to 2024-11-27
News headlines	28,708 AAPL headlines	1,140 news-bearing trading days
Model features	31 stationary features	Market, cyclic time and sentiment candidates
Final model	Logistic regression	Market-only, C=0.1, balanced classes

1. Problem Definition and Scope

The modeling question was deliberately narrow: using information available on trading day t , predict whether AAPL's closing price on trading day $t+1$ will be higher than the close on day t .

Target value	Definition
1 - Up	Next_Close > Close
0 - Down	Next_Close <= Close

Project objectives

• Collect and clean historical AAPL market data and financial-news headlines.

• Classify headlines as negative, neutral or positive and aggregate signals by trading day.

• Engineer technical, lagged-return, volatility, cyclic-time and sentiment features without using future values.

• Compare logistic regression, random forest, XGBoost and LightGBM under chronological validation.

• Explain feature influence using SHAP and communicate results through a deployed dashboard.

Evaluation discipline

Rows were never randomly shuffled across time. Training, validation and test periods were separated chronologically, and the final test set was used once after the configuration was locked. This design reduces look-ahead bias and better approximates deployment on future market observations.

This is an educational analytics system. The output is not financial advice, does not estimate transaction costs or expected returns, and should not be interpreted as a trading recommendation.

2. Data Collection, Cleaning and Integration

Component	Clean result	Role in project
AAPL market history	2,908 feature-ready rows	OHLCV data, target and technical indicators
Labeled financial headlines	4,835 unique rows	Training data for three-class sentiment classifier
Historical AAPL headlines	28,708 scored headlines	Daily sentiment and news-volume features
Unified modeling table	2,210 rows x 41 columns	Chronological classification dataset

Cleaning and alignment decisions

- Standardized dates, numeric types and column names; removed duplicates; verified missing-value counts.
- Defined the target with the next trading day's closing price and removed the final row without a future close.
- Mapped weekend and holiday headlines to the next available trading day before aggregation.
- Filled sentiment features with zero on no-news days and added a News_Available indicator instead of deleting those rows.
- Restricted the unified table to the historical news overlap: 19 Feb 2016 through 27 Nov 2024.

Split	Rows	Date range	Purpose
Training	1,547	2016-02-19 to 2022-04-08	Fit candidate models
Validation	331	2022-04-11 to 2023-08-04	Select features and configuration
Test	332	2023-08-07 to 2024-11-27	Final locked evaluation

3. Financial-News Sentiment Pipeline

A TF-IDF vectorizer and class-weighted logistic regression classifier were trained on labeled financial headlines. The model predicted three labels - negative, neutral and positive - and produced class probabilities for every AAPL headline.

Evaluation	Accuracy	Macro F1	Details
Validation	75.45%	70.22%	725 labeled headlines
Final test	80.03%	76.87%	726 labeled headlines

Final sentiment-class F1-scores were 75.13% for negative, 85.75% for neutral and 69.72% for positive. The saved pipeline combines preprocessing, TF-IDF vocabulary and the classifier so that the dashboard can analyze a new headline consistently.

Daily aggregation

Feature family	Examples
Probability means	Daily_Negative_Score, Daily_Neutral_Score, Daily_Positive_Score
Directional ratios	Positive_News_Ratio, Neutral_News_Ratio, Negative_News_Ratio
Volume and availability	News_Count, News_Count_Log, News_Available
Dispersion	Sentiment_Volatility

A total of 28,708 headlines were scored: 20,824 neutral, 6,201 positive and 1,683 negative. After calendar-to-trading-day alignment, 1,140 trading days contained news and 1,070 did not.

4. Feature Engineering

Feature group	Representative variables	Purpose
Price and return	Daily_Return, Price_Range, Open_Close_Change	Measure same-day movement and intraday spread
Trend	SMA/EMA relations, MACD_Percentage	Represent trend direction and momentum
Oscillator	RSI_Normalized	Capture relative strength on a stable scale
Volatility	Volatility_10, BB_Position, BB_Width	Represent risk and position within Bollinger Bands
Lags	Return_Lag_1, Return_Lag_2, Return_Lag_3	Expose recent sequential behavior
Calendar	Month_Sin, Month_Cos, Day_Sin, Day_Cos	Encode periodic time without ordinal discontinuities
Sentiment	Daily scores, ratios, counts, volatility	Represent news tone, volume and dispersion

Raw price-level features such as Open, High, Low and Close can make a model depend on non-stationary scale rather than repeatable relationships. The final comparison therefore used 31 stationary or normalized features. Date, adjusted close, next close and target were excluded from predictors.

All rolling and lagged variables were calculated from current or earlier rows. No feature used the next day's closing price; it appeared only in the target-construction step.

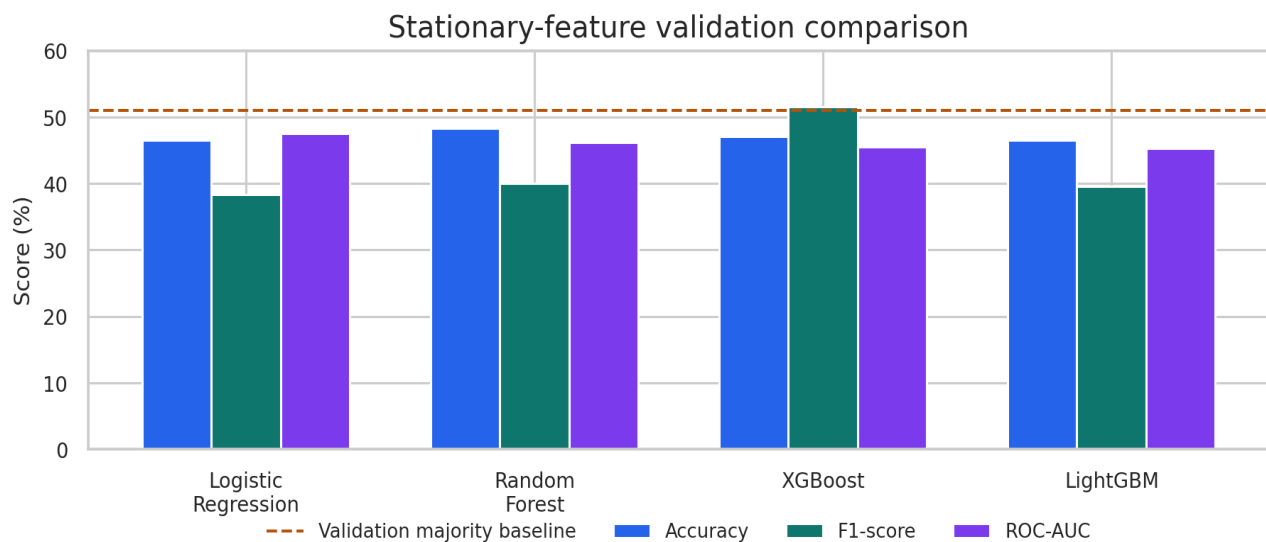
Data integrity checks

- Missing selected-feature values: 0
- Training matrix: 1,547 x 31; validation matrix: 331 x 31; test matrix: 332 x 31
- Training target balance: 832 up days and 715 down days

5. Candidate Models and Chronological Validation

Four supervised classifiers were trained under the same chronological split. Class weighting was used where supported because upward days were moderately more frequent than downward days.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.4653	0.4661	0.3254	0.3833	0.4756
Random Forest	0.4834	0.4914	0.3373	0.4000	0.4620
XGBoost	0.4713	0.4844	0.5503	0.5152	0.4553
LightGBM	0.4653	0.4677	0.3432	0.3959	0.4524



No all-history stationary model exceeded the 51.06% validation majority baseline. This negative result triggered a controlled diagnostic across training windows (all history, from 2019, from 2020 and from 2021), feature groups (market-only versus market plus sentiment), and the four model families.

6. Final Model Selection

The final configuration was selected from validation diagnostics before the test set was evaluated. The best validation accuracy came from a market-only logistic regression trained from 2020 onward.

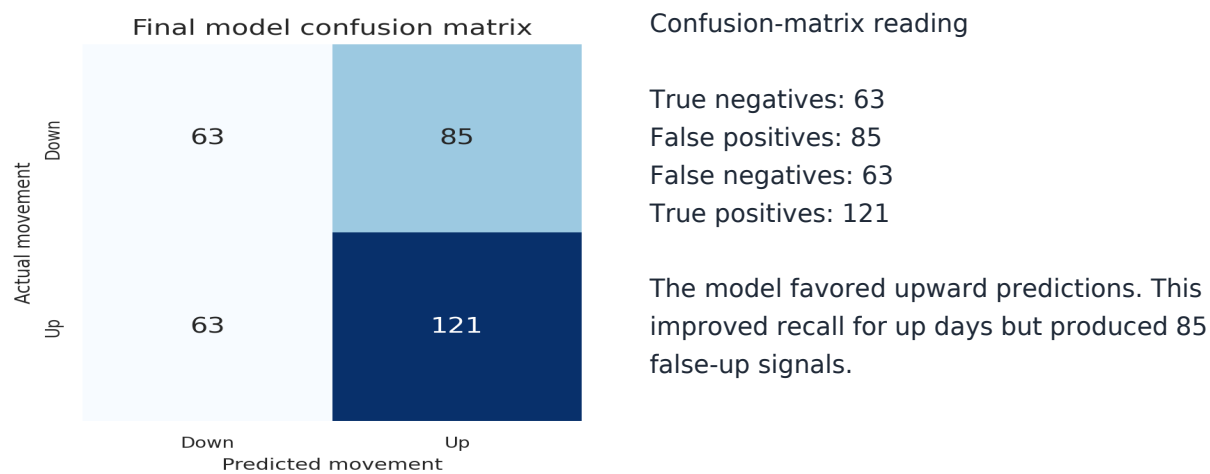
Decision	Locked choice	Reason
Model	Logistic Regression	Best diagnostic validation accuracy; transparent coefficients and probabilities
Training window	From 2020	Reduced exposure to older market regimes
Feature group	Market only	Generalized better than sentiment-enhanced candidate
Regularization	C = 0.1	Stronger regularization for a small, correlated feature set
Class handling	class_weight='balanced'	Counter moderate target imbalance

Diagnostic validation accuracy	52.27%
Diagnostic validation F1-score	57.53%
Diagnostic validation ROC-AUC	48.55%
Training rows in selected window	573

After selection, the model was refit on the available pre-test data from the chosen window and evaluated once on the 332-row test period. The saved joblib bundle contains the fitted model, ordered feature list, training start date and target definition.

7. Final Held-Out Test Results

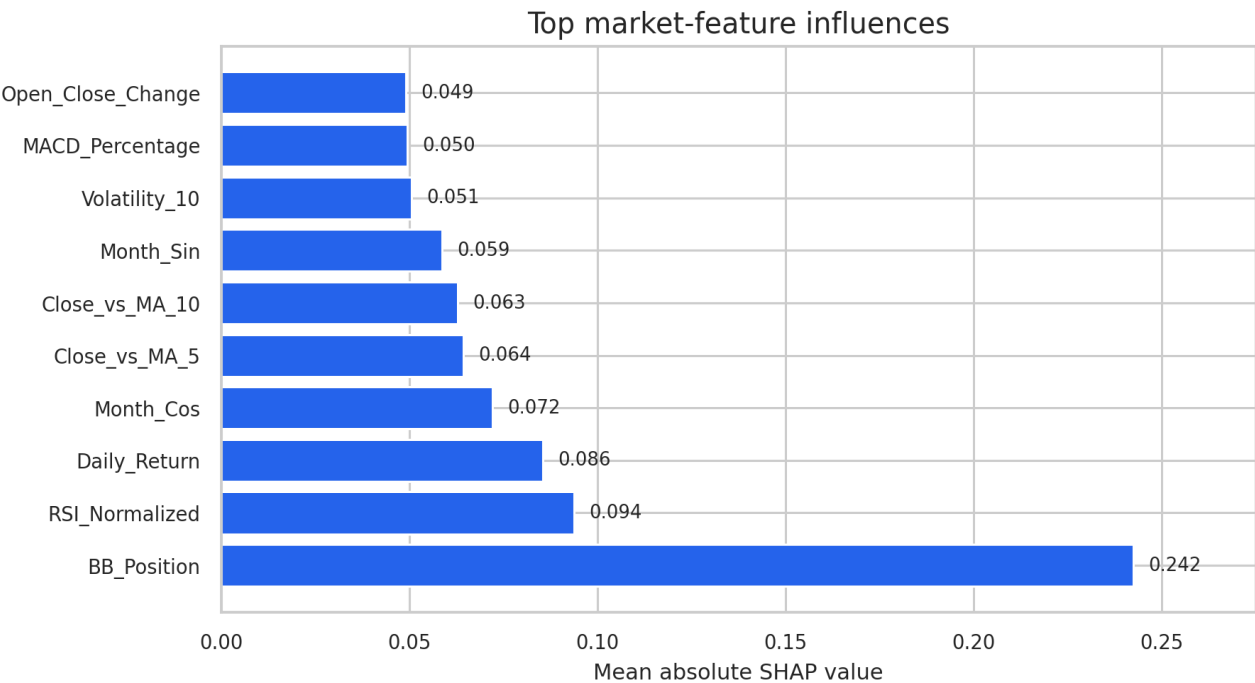
Metric	Score	Interpretation
Majority baseline	0.5542	Always predicting the more frequent test class
Accuracy	0.5542	184 of 332 predictions correct
Precision (Up)	0.5874	58.74% of predicted-up days were actually up
Recall (Up)	0.6576	65.76% of actual-up days were identified
F1-score (Up)	0.6205	Balance of upward precision and recall
ROC-AUC	0.5382	Only modest ranking ability above random chance



The result must be reported candidly: test accuracy did not exceed the majority baseline. ROC-AUC was slightly above 0.50, so the model captured a weak probability-ranking signal, but not enough evidence for reliable trading use.

8. Explainability with SHAP

SHAP values were calculated for the final market-only logistic model. Mean absolute SHAP values rank features by average impact magnitude; they do not by themselves establish causality.

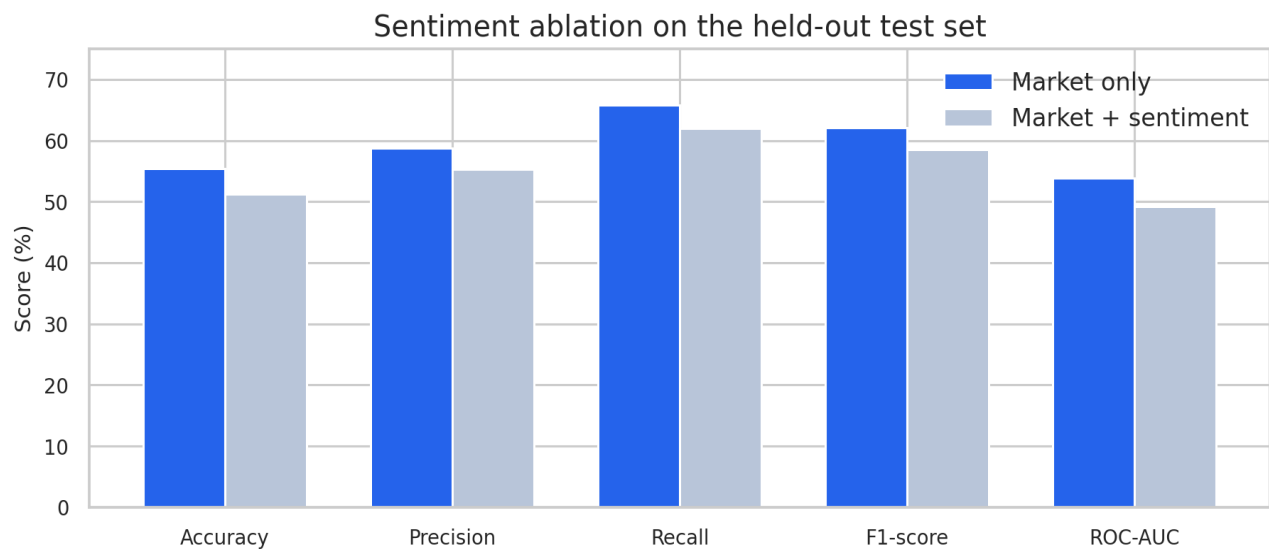


Bollinger Band position dominated average model influence, followed by normalized RSI and daily return. Cyclic month encodings and deviations from short moving averages also contributed. This ranking is consistent with a model driven primarily by normalized price location, momentum and short-term trend relationships.

Direction must be interpreted from the full SHAP distribution or individual observations. A large absolute value indicates strong influence, while the sign determines whether the feature moves the log-odds toward an upward or downward prediction.

9. What News Sentiment Contributed

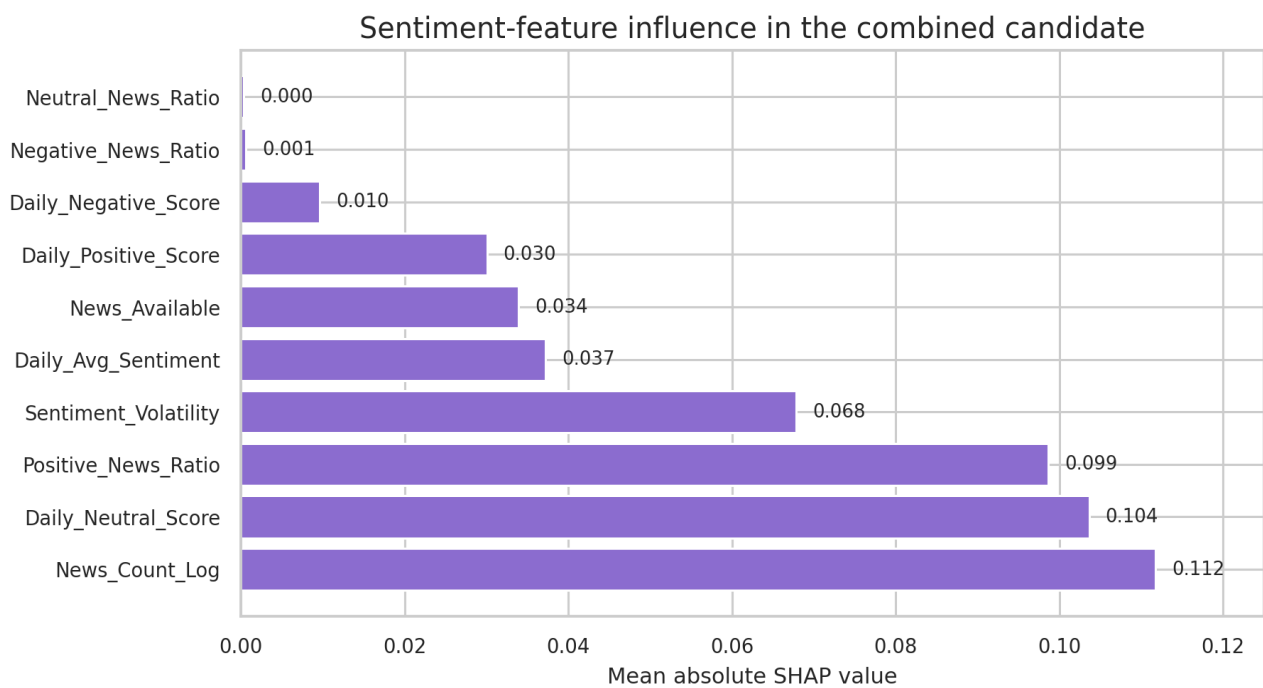
Sentiment features were meaningful inside the combined candidate model, but they did not improve out-of-sample prediction. The ablation compared the locked market-only approach against a market-plus-sentiment candidate on the same test period.



Feature set	Accuracy	Precision	Recall	F1	ROC-AUC
Market only	0.5542	0.5874	0.6576	0.6205	0.5382
Market + sentiment	0.5120	0.5534	0.6196	0.5846	0.4920

Adding sentiment reduced accuracy by 4.22 percentage points and lowered ROC-AUC below 0.50. Therefore, sentiment was retained as a research and explainability component, but excluded from the final predictive model.

10. Sentiment-Feature Influence in the Combined Candidate

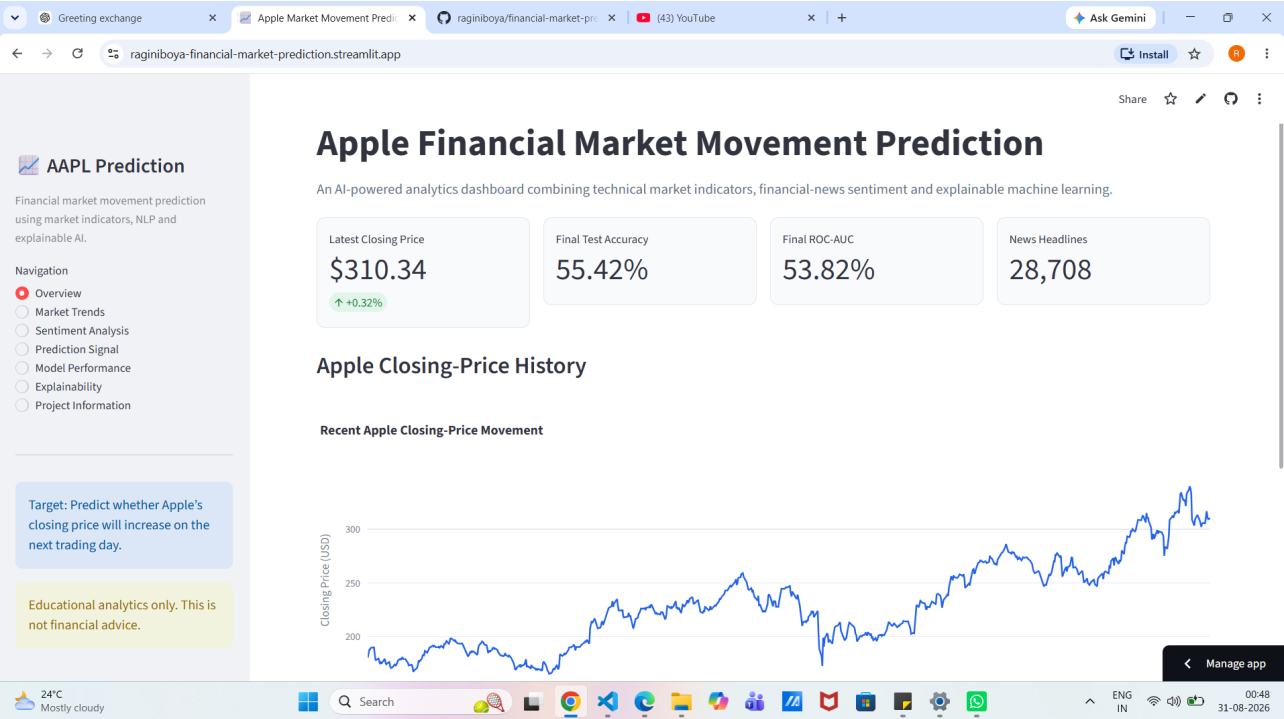


Sentiment variables represented 30.31% of the combined candidate's total mean absolute SHAP importance. News volume, neutral-score level and positive-news ratio were the three strongest sentiment components.

Importance is not the same as usefulness. A feature can strongly change predictions and still reduce generalization if its relationship with the target is unstable, noisy, delayed or regime-dependent. The ablation result is therefore more important for final feature selection than the within-model importance share.

11. Streamlit Application and Deployment

The final Streamlit dashboard exposes the full analytical story rather than only a single prediction. It loads the saved market model, sentiment pipeline, local processed datasets, evaluation reports and generated figures.



Page	Purpose
Overview	Latest AAPL close, project metrics, price history and summary
Market Trends	Price, return, technical-indicator and volume exploration
Sentiment Analysis	Historical sentiment trends plus an interactive headline tester
Prediction Signal	Next-day direction, class probabilities and latest feature values
Model Performance	Metrics, confusion matrix, ROC curve, comparisons and test predictions
Explainability	Market SHAP ranking and sentiment-impact evidence

Live: <https://raginiboya-financial-market-prediction.streamlit.app/>

12. Limitations, Risk and Future Work

Limitation	Why it matters	Recommended next step
Weak predictive edge	Accuracy matched the majority baseline and AUC was 0.5362	Use walk-forward validation and longer out-of-time monitoring
Single asset	AAPL-specific relationships may not generalize	Evaluate multiple equities, sectors and market regimes
Headline-only NLP	Titles omit article context and may repeat symbols	Disambiguate by event and test finance-domain transformers
Daily aggregation	Intraday publication timing is lost	Use timestamp-aware alignment and market-session cutoffs
No trading simulation	Classification metrics do not include costs or risks	Backtest thresholded signals with fees, slippage and risk limits
Regime sensitivity	Feature relationships can change after structural events	Add drift tests, rolling retraining and calibration monitoring

A responsible next version should optimize for stable out-of-time performance rather than in-sample fit. It should also evaluate probability calibration and economic utility, not only classification accuracy.

13. Conclusion and Reproducibility

The project successfully delivered the complete technical pipeline: data collection, preprocessing, sentiment modeling, daily aggregation, technical feature engineering, four-model comparison, chronological validation, final held-out evaluation, SHAP interpretation, saved model artifacts, GitHub source control and a deployed Streamlit application.

Its central analytical conclusion is nuanced. Technical market features produced a small amount of directional signal, while the tested sentiment features changed predictions substantially but reduced out-of-sample performance. The correct response was to exclude sentiment from the final model and report the weak predictive performance transparently.

Deliverable	Status	Location
Data preprocessing notebook	Complete	notebooks/01_data_collection_preprocessing.ipynb
Sentiment pipeline notebook	Complete	notebooks/02_sentiment_analysis_feature_engineering.ipynb
Model evaluation notebook	Complete	notebooks/03_model_training_evaluation.ipynb
Saved prediction model	Complete	models/aapl_movement_logistic_model.joblib
Saved sentiment pipeline	Complete	models/financial_sentiment_pipeline.joblib
Evaluation outputs	Complete	reports/ and outputs/figures/
Streamlit dashboard	Deployed	dashboard/app.py and live URL
Source repository	Published	github.com/raginiboya/financial-market-prediction

Project references

Market data: Yahoo Finance, accessed through the yfinance Python package. Labeled sentiment corpus: Financial PhraseBank-style Kaggle dataset. Historical Apple news: Kaggle Apple news dataset. Model and evaluation evidence: project notebooks and generated CSV/PNG outputs in the public repository.

GitHub: <https://github.com/raginiboya/financial-market-prediction>

Live app: <https://raginiboya-financial-market-prediction.streamlit.app/>